Jargon Buster

Graphics library: A graphics library is a collection of functions that control the output video. These functions can be linked into a program or games to achieve tasks such as initialising video memory, displaying images on the screen, changing colours, etc.
Frame buffer:A frame buffer is the part of memory on a display card that stores the information about what should be displayed on the screen. For example, after rendering video from a game, the graphics card stores every rendered screen in the frame buffer, which is then sent to the monitor or display device when requested. Think of it like and Outbox in your mail client, where mails are stored while they are in the process of being sent!
System event handler: An event handler is a command that's associated with an object. Say, for example, a button on a Web page: you specify a code that basically says, "Do X when this button is clicked". This code is the event handler. Within your system, the same principle is applied for all commands received by a program, the operating system or even the hardware.
Rendering: Rendering is simply the changing of digital information to a human viewable output. Everything from the text you type to the games you play needs to be rendered from ones and zeroes to viewable content. The word rendering, however, is mostly used when describing 3D applications or games, where it means putting together a lot of images to form video.
Pixel shading: Current generation graphics cards have programmable pixel shaders that enable you to calculate the exact colour and texture effect for every pixel displayed on a screen. Even at a resolution of 1024x768 @ 60 Hz, that's 4,71,85,920 computations per second! Using an example from nVidia's site: "Characters now have facial hair and blemishes, golf balls have dimples, a red chair gains a subtle leather look, and wood exhibits texture and grain."
Vertex shading: A vertex is a point on a shape where edges meet. So, a triangle has three vertices—each corner. A vertex shader applies affects and textures to a variety of predefined vertices in an image. This is what helps you get realistic looking waves, because they're not just lines or curves coming towards you, but also have a 3D depth. This is also how games can use lighting effects to determine how to distort a shadow of a moving object, or show the distortion of light coming from around a corner!
Graphics engine: This is a software that takes commands from a program or game and creates images and text, which are then channelled to the graphics driver and hardware.
Raster images: This is an image that's made up of a collection of dots. Bitmap images are raster images and are meant to be displayed on a screen.
Vector images: A vector image is an image that's produced on-the-fly by using mathematical formulas to draw it. Vector images can be scaled up or down without losing any quality, because a graphics engine always has to create the image before being able to display it.

systems, one can be sure that both will continue to co-exist. What might change is the popularity of each API.

Since DirectX is a Microsoft-proprietary technology, a lot of game developers have to use OpenGL in some form or the other in order to be able to release games for the console segment, specifically Sony's PlayStation series. However, with the increase in popularity of the Microsoft Xbox series, game developers can now choose to stick to DirectX and have a presence in the console market.

The future, in terms of PC gaming, might pose problems for OpenGL though. Especially important is the soon-to-be-released next generation OS from Microsoft - Vista. The problem is that Vista uses a 3D graphics engine even to display its regular Desktop windows and a lot of other enhanced effects. It's obvious that Vista, by default, will use DirectX to do its work for it. Actually, Microsoft is calling the new graphics engine Windows Graphics Foundation (WGF), which will perhaps just be DirectX 9.0c renamed. However, there are also claims that Vista will finally ship with WGF 2.0, or DirectX 10 if you like!

Now comes the terrible news for OpenGL: Vista will supposedly run Microsoft's own version of OpenGL (1.4 is the popular speculation) in a layer above WGF (or DirectX). So experts are expecting at least a 50 per cent drop in performance with games written for OpenGL. Though none of this is confirmed, because Vista still hasn't been released yet, it might see game developers such as id Software being forced to code their games on DirectX as well as OpenGL.

Vistas Of Tomorrow

You should have gotten a good idea about what 3D APIs are and a basic idea about how they work. The main concern for most developers today is whether their OpenGL skills will cease to be an asset in the near future. Alas, this is a question that can only be answered once Vista is released later this year.

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Another interesting development is that Microsoft wants to improve the way games are installed on PCs. If we go by popular online hearsay, the WGF will enable us to install games and start playing them in a matter of seconds, as compared to the many minutes it takes to just install games, even on the highest-end PCs. And, unlike with Windows XP, Vista will pass on graphics processing to the GPU directly, without ever disturbing the CPU.

Also, Vista is capable of unloading everything that isn't necessary while playing a game, which includes many services and even the desktop environment—like end-tasking Explorer.exe and then launching a game to prevent wastage of resources. Vista is also touted to come with support for Pixel and Vertex Shader 4.0 (we're barely past version 2.0 currently) so as to future-proof its graphics API.

Whatever the outcome of the API wars, as PCs become more powerful, games get heavier than ever, and our expectations from them increase, some changes will be required within APIs as well as the OSes. Vista seems to bring about such changes, and its WGF standard might change the way all programs are developed, not just games! This is because if Vista becomes the OS of choice (which it probably will), the interface design of every single program will have to be done using raster images instead of the vector images used today! This means that designing even a simple program tomorrow might be more akin to what game designers do today! Whatever the outcome, be prepared for a lot better looking programs, games, and the look-and-feel of Windows as a whole. ■

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